# Extra questions

1. Find the smallest number which ~~or~~ when increased by 17 is exactly divisible by both 468 and 520 [RS 18 19]

Sol: LCM ~~HCF~~ of 468 & 520

$468 = 3 \times 3 \times 2 \times 2 \times 13$
$520 = 2 \times 2 \times 13 \times 5 \times 2$

$LCM = 3^2 \times 2^3 \times 13 \times 5$
$40 \times 117 = \underline{4680}$

The no which when increased by 17 is exactly divisible by both 468 & 520 $= 4680 - 17$

$= \underline{4663}$

2. Find the greatest number of four digits which is exactly divisible by 15, 24 and 36

⇒ Sol: LCM of 15, 24 & 36

$15 = 5 \times 3$
$24 = 2 \times 2 \times 2 \times 3$
$36 = 2 \times 2 \times \cancel{9}\ 3 \times 3$

$5 \times 3$
$2^3 \times 3$
$2^2 \times 3^2$

$LCM = 2^3 \times 3^2 \times 5$
$8 \times 9 \times 5$
$\Rightarrow 360$

Largest Four digit no = 9999

9999/360, leaves 27 as remainder

$360 \times 27 = 9720$

3. In a seminar, the number of participants in Hindi, English and maths are 60, 84, 108, respec ...... find the number of each rooms required

HCF of 60, 84 & 108

$60 = 2 \times 2 \times 3 \times 5$
$84 = 2 \times 2 \times 3 \times 7$
$108 = 2 \times 2 \times 3 \times$

$HCF = 2^2 \times 3$
12

Rooms required for Hindi participant
$= \frac{60}{12} = 5$ rooms

" " for English "
$= \frac{84}{12} = 7$

" " for Maths "
$= \frac{108}{12} = 9$

Total number of rooms =
5 + 9 + 7 = 21 rooms

4. Three sets of English, mathematics and science books containing 336, 240, and 96 books respectively have to be stacked in such a way that all the books are ..... How many stacks will be there [1B 26]

HCF of 336, 240 and 96

$336 = 2^4 \times 3 \times 7$
$240 = 2^5 \times 3$
$96 = 2^4 \times 3 \times 5$

$HCF = 2^4 \times 3 = 48$

No of Stacks for Mathematics

$\Rightarrow \frac{336}{49} = 7$ stacks

" for English

$= \frac{240}{49} = 5$

" for Science

$\frac{96}{49} = 2$

Total no. of stacks $\Rightarrow 7+5+2$

$= 14$

5. Three pieces of timber 42m, 49m and 63m long have to be divided into planks of the same length. What is the greatest possible length of each plank? How many planks are formed.

HCF of 42, 49 & 63

$42 = 2 \times 7 \times 3$

$49 = 7 \times 7$

$63 = 3 \times 7 \times 3$



HCF $= 7 \times 1$

$\Rightarrow 7m$

No of planks made from a plank of 42m $= \frac{42}{7} = 6$ planks

" " $\frac{49}{7}$

$= 7$ Planks

" " " $\frac{63}{7} = 9$ Planks

Total number of Planks

$\Rightarrow 7+6+9 = 22$ planks